ICT DPTE PART 1 - provides knowledge

Diploma in primary teacher education (Meru Teachers' Training College)

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INFORMATION AND COMMUNICATION TECHNOLOGY

Information Technology (IT) is a technology which uses computers to gather, process, store, retrieve and protect, information. Today, it is common to use the term Information and communications technology (ICT) because it is unimaginable to work on a computer which is not connected to the network. **Information Communications Technology** (ICT) therefore refers to the use of computers that are connected to the network to gather, process, store, retrieve, protect, and transfer information.

A **computer** is an information-processing machine. It may also be defined as a device that works under the control of stored programs automatically accepting, storing and processing data to produce information that is the result of that processing.

Data refers to raw facts which have not been processed in to meaningful information. **Processing** is transforming data in to meaningful information. Usually, Processing takes place in the CPU (Central processing unit).**Information** refers to meaningful output produced by a computer system.

Why use computers?

Use of computers has become a necessity in many fields. Computers have revolutionized the way businesses are conducted. This is due to the advantages that computer systems offer over manual systems.

The advantages include:

1. *Speed* – Computers have higher processing speeds than other means of processing, measured as number of instructions executed per second.
2. *Accuracy* – Computers are not prone to errors. So long as the programs are correct, they will always give correct output.
3. *Consistency* – Given the same data and the same instructions computers will produce exactly the same answer every time that particular process is repeated.
4. *Reliability* – Computer systems are built with fault tolerance features, meaning that failure of one of the components does not necessarily lead to failure of the whole system.
5. *Memory capability* – A computer has the ability to store and access large volumes of data.
6. *Automation:* A lot of tasks can be automated saving a lot of time. For example, instead of manually calculating some values like mean, median of a large dataset, we just use excel.
7. *Data retrieval:* With one touch of a button, finding documents becomes easy. Thus it saves a lot of time.

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Disadvantages of using computers

1. **Unemployment:** Computers can perform many tasks automatically; this reduces the need for people and increases unemployment in the society.

2. **Health issues**: Prolonged use of computers can negatively affect your health. For example, if you are constantly working on the computer, your eyes will become dry, which can result in eye strain, headache, etc.

3. **Cybercrime**: Cybercrime refers to criminal activities carried out by means of computers or the internet, e.g. hacking, cracking, tapping etc.

4. **Addiction and time wastage**: Some people use computers for entertainment purposes like playing games, and it lasts an extended period, which has negative impacts. A much younger generation is currently spending more time and energy on the computer.

5. **Dependence on electricity**: A computer needs electricity to process and work; without electricity, a computer is no more than a metal box

6. **High cost –** Computers are expensive. Even the foremost affordable computers are still very expensive for the typical person.

7. **Increases waste and impacts the environment –** With the speed that computers and other electronics get replaced, all of the old devices that get thrown away have a big negative impact on the environment

Computer application areas

Some of the areas that computers are used include:

– **Communication** –Computers have enhanced communication through email communication, electronic data interchange, electronic funds transfer, Internet etc.

– **Banking** – the banking sector has incorporated computer systems in such areas as fund transfers, customer relations, automated teller machines, home banking, and online banking.

– **Organizational management** –Computers are used in organization for transaction processing, managerial control as well as decision-support.

– **Education**– computers incorporate databases of information that are useful in organizing and distributing educational resources. Such E-learning and distributed classrooms have enabled the teaching industry to have a global reach to the students.

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- **Management of information materials**- The Internet has massive reference material on virtually every learning area. Computer systems have enabled the efficient running of libraries for information storage and retrieval.
- **Home appliances** – computers (especially embedded computers or microprocessors) are included in household items for reasons of economy and efficiency of such items. Major appliances such as microwave ovens, clothes washers, refrigerators and sewing machines are making regular use of microprocessors.
- **Health care and medicine** – computers have played an important role in the growth and improvement of health care that the use of computers in medicine has become a medical specialty in itself. Computers are used in such areas as maintenance of patient records, medical insurance systems, medical diagnosis, and patient monitoring.
- **Reservation systems** – guest booking, accommodation and bills accounting using computers in hotels have made the process to be more efficient and faster. Airline computer reservation systems have also enhanced and streamlined air travel across major airlines. Major players in the industry have also adopted online reservation systems.
- **Entertainment** – use of computers in the entertainment industry has increased tremendously over the years. Computers enable high-quality storage of motion pictures and music files using high-speed and efficient digital storage devices such as CDs, VCDs and DVDs. The Internet is also a great source of entertainment resources. Computer games have also become a major source of entertainment.
- **Retailing** – computers are used in point of sale systems and credit card payment systems as well as stock inventories.

Components/Elements of a computer system

A system is a collection of components that work together to achieve some common goal. Each system is made up of small units or components known as subsystems. These components are:

1. Hardware – These are the physical and tangible parts of a computer system E.g. Keyboard, mouse, etc
2. Software – These are Programs or instructions that tell the computer what to do and how to do it.
3. Data/Information – Data refers to raw facts which have not been processed in to meaningful information.
4. Personnel – These are people using the computer system. E.g. End users, programmers, system administrators etc.
5. Procedures **–** instructions that helps people to use the computer system.

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Data processing Lifecycle

These are the stages that data is passed through during processing. Also known as activities or operations of computer system

1. **Input** – at this stage data from the environment is entered in to the information system for processing. The input hardware devices here may include keyboard, mouse, joystick, etc
2. **Processing** – This is transforming data in to meaningful information. Usually, Processing takes place in the CPU (Central processing unit)
3. **Output** – meaningful information produced by information systems. Output hardware devices may include; monitors printers, speakers, projectors etc.
4. **Feedback** – Returning the information to the appropriate people in the organization.
5. **Storage** – writing data and information in to a secondary storage media. E.g. hard disk, flash disk, memory card, etc

The Central processing unit (CPU)

The central processing unit is the control center of the entire computer system. It accepts data from various input devices, processes this data in to information, and sends this information to an output device such as a monitor or a printer. The central processing unit is composed of two major components namely;

The central processing unit is composed of three major components namely;

Arithmetic logic unit (ALU): The arithmetic logic unit is the computer's calculator. It performs all the arithmetic operations, in addition to decision making and editing functions. The arithmetic operations include operations such as additions, subtractions, divisions and multiplications. Logic operations include arrangement and rearrangements, matching, comparing, etc.

Control unit (CU): The control unit controls, and coordinates all operations occurring within the central processing unit. It controls program execution. The arithmetic logic unit functions under direct control of the control unit (CU).

Registers: Temporally storage areas used by the arithmetic logic unit and the control unit When performing their operations.

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Computer Hardware

Computer hardware refers to the physical parts or components of a computer such as monitor, keyboard, Computer data storage, hard drive disk, mouse, system unit (graphic cards, sound cards, memory, motherboard and chips), etc. all of which are physical objects that you can actually touch.

Computer Hardware components are broadly divided into four types:

1. **Input Devices:** Hardware used to input data into computer system e.g. keyboard, mouse, ec
2. **Output Devices:** Hardware used to output processed data from computer system e.g. Monitor, printer, etc
3. **Storage Devices:** Hardware used to hold data, information and instructions for future use. e.g. hard disk ,flash disk, CD, etc
4. **Communication devices:** Allows computers to send and receive data and information to and from other computers.
5. **System Unit:** Hardware present in System Unit Like CPU , Motherboard, Graphics Card, Data Bus, System Cables, Ports etc

System Unit

The system unit refers to the body of the computer containing the electronic parts that operate the computer, the connections to the other components, and the disk drives. It is important to note that the **central processing unit** (CPU) is contained within the system unit. Among other components that contained within the system unit is the **mother board, Processor, RAM, ROM, BIOS, disk drives, floppy disk, hard disk** and **power supply.**

1. **Motherboard:** The motherboard is the main circuit board of the system unit

2. **Processor (CPU):** The central processing unit is the control center of the entire computer system. It accepts data from various input devices, processes this data in to information, and sends this information to an output device such as a monitor or a printer. The central processing unit is composed of two major components namely;

– **Arithmetic logic unit (ALU):** The arithmetic logic unit is the computer's calculator. It performs all the arithmetic operations, in addition to decision making and editing functions.

– **Control unit (CU):** The control unit controls, and coordinates all operations occurring within the central processing unit. It controls program execution.

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3. **Random access memory (RAM)** is an internal chip where data is temporarily stored when running applications. This memory can be written to and read from. Since its contents are lost when power to the computer is turned off, it is often referred to as a '**volatile**' or '**temporary**' **memory.**
4. **Read only memory (ROM)** is a memory used to store information that needs to be permanent. It is often used to contain, for example, configuration data for a computer system. These chips cannot be altered and can only be read from (hence their name). One of the main advantages is that the information stored on the ROM chip is not lost even when power is turned off to the computer. They are often referred to as '**non-volatile' memories.**

5. **Bios** – This is the program that starts the computer system. The bios program controls the system during initialization process.

6. **Disk Drives** - Disk drives are devices inside the system unit that read and write information on a disk. Data stored on a disk is not lost when the computer is turned off.

7. **Hard Disk–**The hard disks is a secondary storage device for large capacity storage of data. They range in capacity from 20 megabytes and larger.

8. **A power supply unit (PSU)** converts alternating current (AC) electric power to low-voltage DC power for the internal components of the computer.

9. **CMOS (**complementary metal oxide semiconductor**):** a non-volatile chip stores the date, time and system configuration

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Computer Software

Computer Software refers to programs or applications that tell the computer hardware what to do and how to do it.

Software is broadly divided into two types:

1. **System software**: Softwares designed to meet computer's needs. These Softwares manages and controls computer hardware so that application softwares can perform their task. e.g. Operating System (OS) and utility programs.

2. **Application software**: Softwares designed to meet user's needs. They are designed to enable the end-user to perform specific productive tasks. e.g. MS Word for word processing or Photoshop for image manipulation.

Hardware and Software have a symbiotic relationship, this means that without software hardware is very limited; and without hardware, software wouldn't be able to run at all. They need each other to fulfill their potential.

Differences between system software and Application software

System software	Application Software
– System software is designed to meet computer's needs.	– Application software is designed to meet user's needs
– System software gets installed when OS is installed on the computer	– Application software gets installed according to the requirements of the user
– System software can run independently without application software	– Application software cannot run independently without the presence of the system software.
– Generally, users do not interact with system software as it works in the background	– Users interact with application software while doing different activities.
– A computer may not require more than one type of system software	– there may be a number of application software programs installed on the computer at the same time

Application software can further be classified in to two;

i. Bespoke software
ii. Of-the-shelf software

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Bespoke Software/Custom software/ Unique Applications/Proprietary software

Bespoke Software is software designed to suit your company's unique requirements. Bespoke software is written specifically for you, to meet your specific business requirements. They are also known as **in-house programs** or as **tailor made programs.**

Advantages of bespoke software

- The software more closely meets your requirements, as it is developed especially for you.
- Bespoke software is easy to use because it works the way you work.
- It can evolve over time to match your changing requirements.
- You only pay for the features you need.
- If you own the software, you won't have to pay a monthly license or similar ongoing cost.
- It may give you a competitive advantage, as your competitors won't have the same software.
- Bespoke software precisely matches your working practices, resulting in improved efficiency, less supervision, fewer errors
- Bespoke applications increase productivity, and reduce costs by automating repetitive tasks.

Disadvantages of bespoke software

- Highcost of building the software.
- It takes longer to get the system set up, as you have to wait for it to be developed.
- It is extremely difficult get a professional software developer to develop the software
- If you lose the source code for your application your business could be exposed to significant risk.

Off The Shelf software/Packaged software/Commercial software

Off the shelf software is designed for a wide range of businesses and may not be ideally suited to your specific business. They are also known as application software or packages.

Advantages

1. They are less expensive to acquire compared to developing the software.
2. It often has more functionality than bespoke software, because the developers try to satisfy more use-cases.
3. If it is a commonly used package, users and I.T. staff may already be familiar with it, saving on learning time and training costs.
4. Books and online help are usually available to help you learn or resolve problems.
5. They are usually accompanied by proper documentation which helps users learning the software.
6. The packages are generally portable, i.e. they can be run by different models of computers.

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Disadvantages of off-the-shelf software/Application software/Packages

1. It is not easy to modify the packages.Users must depend on suppliers in case of changing the software.
2. The packages may not be compatible to some operating systems e.g. Ubuntu.
3. The programs may not be suitable to solve some specific problems.
4. Programmers are needed to install the packages that fit the organizational needs.
5. It is unlikely you will find ready-made software that does everything you would like it to.
6. As your requirements change, the software may not be able to change to keep up – this can lead to a costly switch to a new package.
7. The same system is available to your competition and gaining any lasting advantage from it is unlikely.

Operating System (OS)

An operating system (OS) is system software that manages computer hardware, software resources, and provides common services for computer programs. Common examples of operating systems include: Windows, linux, unix, MS DOS, and Android

Functions of an operating system

1. Booting the computer
2. The OS manages all the hardware and peripheral devices of your computer. eg mouse, keyboard
3. provides the interface between the user and the computer, e.g. command line, graphical user interface (GUI)
4. Handles system resources such as computer's memory and sharing of the central processing unit (CPU) time by various applications or peripheral devices
5. Provides file management which refers to the way that the operating system manipulates, stores, retrieves and saves data.
6. Memory allocation: The OS deals with the transfer of programs in and out of memory
7. maintains security and access rights of users
8. Error reporting; issues simple error messages when an error occurs during program execution
9. **Processor management:** At times, several tasks may require processing, hence creating competition. To do this, the OS arranges the tasks according to priority for processing.

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TYPES/CLASSIFICATION OF OPERATING SYSTEMS (OS)

The OS determines determine the type of processing that a computer system is able to perform, since it controls the allocation & use of the computer resources. Operating systems can be classified according to:

1. Number of tasks that the system can perform concurrently.

- **Single-taskingoperating system**. E.g. DOS: A single tasking operating system is an operating system that allows only a single program to run at a time.
- **Multi-tasking operating system**. A multi-taskingoperating system is an operating system that is capable of allowing multiple programs to run at the same time.

2. Number of users the system can support at the same time.

- **Single-user operating systems**. e.g DOS: Single-user operating systems, as opposed to a multi-user operating system, are usable by a single user at a time.
- **Multi-user operating systems**. E.g Linux, Unix, Windows. A multi-user operating system allows for multiple users to use the same computer at the same time and different times.

3. Number of processors it can support

- **Single processing operating system:**An operating system capable of supporting and utilizing more than one computer processor.
- **Multi-processing operating system:**An operating system capable of supporting and utilizing more than one computer processor.

4. Human Computer interface (**HCI)** (i.e., how the user & the computer interact).

- **Command line interface:** Allows the user & the computer to interact using commands
- **Menu driven interface:**This type of interface provides the user with a list of program commands displayed on the screen to choose from
- **Graphical user interface (GUI):** Allows the user & computer to interact by clicking on **Icons using the mouse**

5. **Real time OS:** Responds to input instantly. General-purpose operating systems, such as DOS and UNIX, are not real-time.

6. Distributed OS - A distributed operating system manages a group of independent computers and makes them as a single computer.

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Factors to consider when choosing an operating system

The following factors should be considered when choosing an operating system for a computer;

a. Hardware configuration of the computer, e.g., RAM memory size, hard disk capacity, type of processor, etc.
b. User needs (requirements), i.e., the applications intended for the computer.
c. User friendliness or Human computer interface, i.e., is it Command line based, Menu-driven or a Graphical user interface?
d. Availability in the market, e.g. Microsoft Windows based OS are very common.
e. Portability.
f. Cost – how expensive the OS is.
g. Reliability, i.e., can it run without *crashing* or *hanging* (stop responding to commands).

User interface

A user interface controls how you enter data and instructions and how information is displayed on the screen

There Are Two User interface provided by Operating System

1. **Graphical User Interface (GUI):** User interact the system through WIMP environment-Windows, Icon, Menus, and Pointers (WIMP). GUIs are small pictures that represent actions, and they can be selected by 'clicking' on them with the mouse. Example of GUI interface Windows & Mac OS

2. **Command line interfaces (CLI):** require a user to type in instructions in order to choose options from menus, open software etc. There are often a number of commands that need to be typed in, for example, to save or load a file. The user therefore has to learn a number of commands just to carry out basic operations. E.g MS DOS

A Utility Program is a type of system software that allows a user to perform maintenance-type tasks. Examples of Utility Programs : Windows Explorer (File/Folder Management), Compression Tool, Anti-Virus Utilities, Disk Defragmentation, Disk Clean, BackUp, WinZip, WinRAR etc

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Evolution of computers

The earliest mechanical aid to calculation was abacus or bread frame. This is said to have originated in china in the seventh century BC. The Romans used pebbles on a grooved board in similar way to the abacus; and the word "calculation" is derived from the Latin 'calculus' meaning a pebble. Therefore the first computer to be developed was a basic adding machine (calculator) for performing only additional functions.

The computers of today are vastly different in appearance and performance as compared to the computers of earlier days. But where did this technology come from and where is it heading? To fully understand the impact of computers on today's world and the promises they hold for the future, it is important to understand the evolution of computers.

COMPUTER GENERATIONS

The term computer generation is applied to different types of computers to help describe the major technological developments in hardware and software. Today, computer technology has evolved through four distinct generations and is currently developing in to the fifth generation. With each new generation, the circuitry has gotten smaller and more advanced than the previous generation before it. As a result of the miniaturization, speed, power, and memory of computers have proportionally increased.

First generation (1944-1958)

The first generation computers had their main memory made up of thousands of vacuum tubes.

Characteristics of the first generation computers

- They were too huge in size thus requiring large space for installation.
- They used vacuum tubes for the main memory.
- They had limited commercial use since they were difficult to program and use.
- They were very expensive to install the hardware.
- These computers could only run one program at a time.
- They consumed a lot of power and produced tremendous amounts of heat hence not reliable for computer circuitry since the components were prone to failures.

Second generation (1959-1963)

In the second generation, the transistors were invented to replace vacuum tubes for computer circuitry. One transistor replaced the equivalent of 40 vacuum tubes. Therefore they were very cheap to produce.

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Characteristics of the second generation computers

- They used transistors unlike the first generation which used vacuum tubes.
- They were smaller than the first generation computers and required smaller space.
- They were cheap to produce compared to the first generation since One transistor replaced the equivalent of 40 vacuum tubes
- They consumed less power and generated less amounts of heat compared to the first generation computers thus less prone to hardware failures.
- They were easier to use and program compared to the first generation computers hence had a wider commercial use.

Third generation (1964 - 1970)

In the third generation, the integrated circuit was invented to replace the transistors for computer circuitry. Integrated circuit means a computer electronic circuit on a silicon chip.

Characteristics of the third generation

- They used the integrated circuits
- They were smaller than the second generation thus requiring smaller space.
- They were more reliable and less prone to hardware failures compared to the second generation computers.
- They had higher main memory capacity than the second generation computers.
- They could support more than one user at the same time.
- They could process several programs at the same time.
- They consumed less power compared to the second generation computers.

Fourth generation (1971 - now)

They used large scale integration (LSI) for computer circuitry and main memory. This generation is characterized by monolithic Integrated circuits (millions of transistors put onto one integrated circuit chip).

Characteristics of the fourth generation computers

- They used large scale integration for main memory and logic circuitry.
- They were more powerful and had faster processing speeds compared to the third generation computers.
- They were general purpose machines.
- They consumed less power compared to the third generation machines.
- They had larger primary and secondary storage compared to the third generation computers.

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Fifth generation (now and in future)

Definitions of what constitutes fifth generation computers do not always agree. Some people think that the new microcomputers with faster operating speeds, greater processing capacity and virtually unlimited memory should be included. Other people believe that fifth generation computers will have circuitry based with gallium arsenide. Gallium arsenide offers a fivefold speed increase and uses only one tenth of the power that silicon uses. Many fifth generation computers will also incorporate hundreds or thousands of processors that operates in parallel i.e. simultaneously.

Characteristics of fifth generation computers

- The ULSI {Ultra Large Scale Integrated Microprocessor} Technology is used in fifth generation computers.
- They use parallel processing.
- Artificial Intelligence is used in these 5th generations of computers, and these characteristics are considered the most acknowledged and utilized.
- These computers are more reliable and portable compared to other generations of computers.
- They are relatively cheaper than their counterparts.
- Use of optical fiber in circuits

Classification of computers

Because of the variety of computers power available, computers are classified as to **purpose, type** and **capacity.**

Purpose

Computers are designed to either **special purpose** or **general purpose** computing devices.

Special purpose computers- are also known as dedicated computers and are computers designed to perform only one major function. eg calculators, Thermometers, etc

General purpose computers **-** are designed to handle a variety of tasks. General purpose computers are also known as hybrid computers. eg PCs, Cell phones, Notebooks, etc

Type

There are two types of computers currently available; **analog and digital**

Analog Computer: The computer that work with natural phenomena and physical values like earthquake measurement, speed of the wind, weight light etc is known as Analog computers. It

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is especially used in scientific work, medical and industrial field. These are special purpose computers. It measures physical values such as temperature or pressure that fall along a continuous scale in temperature or pressure. For example, you can see a system on petrol pump that contains an analog processor and analog device that converts flow of petrol into quantity. Speedometer in cars and your watch are other examples of analog computer

An analog computer is a form of computer that uses *continuous* physical phenomena such as electrical, mechanical, or hydraulic quantities to model the problem being solved.

Digital Computer:

A computer that work with digital value 0 and 1. Where 0 is OFF and 1 is ON. It works with discrete data. Digital computer does not measure the continuous data for continuous output. Most of the electronic system is based on the digital system. Digital computers are very popular for actual computers are very popular for actual computer works like preparation of the report, documentation, billing and other graphical work etc. The entire PC (Personal Computer) used today on different fields are digital computers.

A computer that performs calculations and logical operations with quantities represented as digits, usually in the binary number system

Hybrid computer:

It is the combination of analog and digital computer system. It works with continuous and discrete value. The good qualities of analog and digital computers are combined on this computer and made the hybrid computer. These are used in ICU (Intensive Care Unit) of the hospital, jet planes, and other data analysis terminals. Hybrid computer transfers the data from analog to digital and digital to analog and vice-versa.

Capacity/ Physical size

The capacity of a computer refers to the volume of data a computer system can process. The size of computer is measured by its throughput i.e. the amount of processing that can be put through it in a given amount of time. Based on throughput, computers systems fall in to four categories:

– **Microcomputers -** These are the most common forms of computers in offices today. They are small desktop or laptop systems or personal computers. They are the smallest of all the classes compared to main frames and mini computers.

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- **Minicomputers:** They are medium sized computers used in medium sized businesses e.g. insurance companies, banks etc. Usually they have a higher processing and storage capacity than microcomputers.
- **Mainframe computers:** Generally mainframe computers have greater processing speeds over microcomputers and minicomputers. They are used in large business organizations which handles complex and voluminous data e.g. government offices. It is a very large and expensive computer capable of supporting hundreds, or even thousands, of users simultaneously.

- **Supercomputers:** The fastest and most powerful type of computer. Supercomputers are very expensive and are employed for specialized applications that require immense amounts of mathematical calculations. For example, weather forecasting requires a supercomputer.

Types of Micro Computers/ Personal Computers

- **Computer**: A desktop computer is a personal computer designed for regular use at a single location on or near a desk due to its size and power requirements
- **Laptop Computer**: a portable computer complete with an integrated screen and keyboard. It is generally smaller in size than a desktop computer and larger than a notebook computer. It is practically placed on the user's lap, hence the name.
- **Tablet Computer:** tablet computer, *computer that is intermediate in size between a laptop computer and a smartphone*.
- **Smartphone:** a mobile phone that performs many of the functions of a computer, typically having a touchscreen interface, internet access, and an operating system capable of running downloaded apps.
- **Personal digital assistant (PDA):** A personal digital assistant, usually referred to simply as a PDA or a handheld PC, is a microcomputer that acts chiefly as a digital information manager.
- **Programmable calculator**– pprogrammable calculators are calculators that can automatically carry out a sequence of operations under control of a stored program

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ICT TOOLS

Information Communication Technology tools are digital infrastructures such as; computers, laptops, desktops, data projector, software programs, printers scanners and Interactive teaching box.

Examples of ICTs that may be used as educational tools in the classroom may include: Radio, Television, computer, Internet, mobile learning, whiteboards, etc

Information and Communication Technology consists of various tools and systems that can be exploited by capable and creative teachers to improve teaching and learning situations. Lim and Tay (2003) classification of ICT tools as:

1. **Informative Tools**: Informative tools are applications that provide large amounts of information in various formats such as text, graphics, sound, or video. Examples include; Internet, Network Virtual Drive, Intranet systems, Homepage, E-books, etc.

2. **Communicative Tools:** Communicative tools are systems that allow easy communication between teachers and students or between students outside the physical barrier classroom. It is including e-mail, electronic bulletin boards, chat, teleconference and electronic whiteboard.

3. **Situating tools**: Situating tools is the system that situates users in an environment where they may experience the context and the happening. Examples of such systems include simulation, virtual reality and multi-user domain. Situating tools software tools such as CD-ROM.
4. **Collaborative tools**: A collaboration tool helps people to collaborate. The purpose of a collaboration tool is to support a group of two or more individuals to accomplish a common goal or objective. Internet can be used for many collaborative activities such as meetings, discussions, information dissemination, and other tasks.

5. **Constructive Tools:** constructive tool is a general purpose tool that can be used to manipulate information, construct their own knowledge or visualize students understanding. Construction tools such as Microsoft Word or Powerpoint has a strong impact in the educational environment

6. **Classroom equipments**: Standalone equipments that are used in traditional classroom to facilitate the interaction between teachers and students in different class activities, e.g. Interactive whiteboard, touch screen computer, kiosk, etc

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ICT Tools for teaching and learning

The ICTs are needed at school level for the following activities (Sansanwal, 2009):

- Teaching and learning process
- Diagnostic Testing and Remedial teaching
- Evaluation activities
- Psychological analysis of learners
- Development of reasoning and thinking abilities among students
- Instructional material development

Factors to consider when selecting ICT Tools for learning

1. **Student needs**—taking into account their prior knowledge, level of technological literacy, personal interests, and those other things that make our students special and unique individuals

2. **Compatibility**: When choosing hardware, make sure that any new or replacement components are compatiblewith your existing computer equipment.

3. **Individual or Collaborative?:** Is this tool meant to give students a sense of community or is it meant for individual work? Some digital tools might provide those students at the top of the class a chance to connect with similar students.

4. **Costs and Savings:** The cost of the tool is another factor to consider. Free and cheaper software is easier to access and may have a larger user community. Also, a digital tool may actually end up saving money for schools, teachers and students. So the question may actually be about how much money the tool will save you, rather than cost you.

5. **Familiarity with Technology:** There should be a consideration about how comfortable the student is with digital tools. Think about tools they have used previously and how they compare to the potential selection.

6. **User friendly** – The ICT Tool should be easy to use for various purposes like generating quizzes, evaluating students' performance and so on

Methods of acquiring ICT Tools for use in supporting learning

Schools acquired their computers through a combination of different methods;

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1. Buying/purchasing
2. International donations
3. Corporate/Alumni donation
4. NGO Donation
5. Government donation

PRODUCTIVITY TOOLS

Basic Productivity Tools

Basic productivity tools are computer software programs which allow a user to create specific items quickly and easily as opposed to creating the same items by hand. They help users produce things such as documents, databases, graphs, spreadsheets and presentations. Some of the common productivity tools include:

1. **Word Processing**. A word processor is a program that is used to create, view, edit, and print documents. An example would be Microsoft Word.

2. **Spreadsheets.** A spreadsheet is an application tool that store, organize, and calculate data in tables. An example would be Microsoft Excel.

3. **Presentation program**: a presentation program (also called presentation software) is *a software package used to display information in the form of a slide show*. An example would be Microsoft powerpoint.

4. **Databases**. Databases allow the user to save collections of information in one easily accessible place. This allows the user to find information about a specific topic much faster versus searching the internet. An example of a database would be Microsoft access, Galileo, etc
5. **Graphics suite:** A graphics suite is a group of programs that are used to view, manipulate, and print computer graphics. Popular examples are Corel and Adobe, which includes Photoshop.

Productivity tools enhance a teacher's lesson plan and allow the teacher to take their students beyond textbooks and boring lectures. For example, Excel is a productivity tool that can be very helpful in teaching situation dealing with math and science. Excel possesses the ability to change numerical data and put it in the form of a graph in a matter of seconds. This technique allows the teacher to show the difference in the results of an experiment or survey using a graph or chart. This visual picture is just as important as the numerical data. Some children are

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visual learners and they may not be able to comprehend the difference of the data without seeing it in a chart or graph. On the other hand, PowerPoint enables teachers to make slides of their notes and project them onto a screen so that everyone can copy word for word what the teacher wants them to learn.

Examples of productivity suits

A productivity suite is a set of applications that generally includes at least a word processing, spreadsheet, and presentation graphics program. There are many productivity tools available, both as standalone applications and as part of suites. Here are some of the most popular examples:

- Microsoft Office is one of the most well-known suites, which includes Word, Excel, PowerPoint, OneNote, Outlook and Publisher.
- Google Workspace is a suite that includes Google Docs, Google Sheets, Google Slides and Google Forms.
- Apache OpenOffice is a popular open source suite that includes Writer for word processing, Calc for spreadsheets, Impress for presentations, Draw for graphics and Base for databases.
- Apple iWork is a suite from Apple that includes Pages for word processing, Numbers for spreadsheets and Keynote for presentations.
- Openoffice: Another free office productivity suite. The following are included in this suite: Writer, Calc, Impress, Draw, Base, and Math.

What are the benefits of using productivity software?

There are many benefits to using productivity software, including the following:

- **Increased efficiency.** By automating tasks and providing templates and other tools, productivity software can help users work more quickly and efficiently.
- **Improved accuracy.** Automated features can help reduce errors in data entry and other tasks.
- **Greater flexibility.** Productivity software often provides a range of ways to view and manipulate data, making it easier to find the information you need and tailor it to your specific needs.
- **Enhanced collaboration.** Many productivity applications now include features that make it easy to share documents and work on them together with colleagues, which can improve communication and collaboration.

Disadvantages of using productivity software?

There can also be some disadvantages to using productivity software, including the following:

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- **Increased dependence on technology.** Relying on productivity software can make it difficult to work if there are technical problems or employees don't have access to a computer.
- **IT support issues.** If employees are not familiar with the software, they might need help from IT support staff to use it effectively, which can add to costs.
- **Complexity.** Some productivity applications can be complex and time-consuming to learn, which can offset any efficiency gains.

How to choose the right productivity software

When choosing productivity software, it's important to consider your specific needs and requirements. Below are a few key points to keep in mind when selecting productivity software.

- **The types of tasks you need to do.** Make a list of the tasks you need to be able to do with the software. This will help you narrow down your options.
- **How many features you need.** If you're just using it for personal use, you might not need as many features as if you're using it for business purposes.
- **Your budget.** Productivity suites can be expensive, so it's important to consider how much you're willing to spend. There are also some good free options available.
- **Your level of expertise.** If you're not very familiar with computers, you might want to choose a tool that's relatively easy to use.
- **Compatibility with other software.** Make sure the productivity software you choose is compatible with any other software you're using.

Emerging productivity tools in learning

- **Google Docs**: Google Docs is a free service that allows you to upload documents, edit them, and share with others. Your students need not print their papers to turn in; they can simply upload and share with you.

- **Rescue Time**: Rescue time is an online time tracking and management program for any busy person who wants to get more done with their time. When entering grades into online grade books, it is too easy to flip over to email or Facebook and lose track of time. RescueTime can block those distractions to make it easier for you to stay focused.

- **Virtual assistant:** A virtual assistant is an independent contractor who provides administrative services to clients while operating outside of the client's office. You can sign up with The International Virtual Assistants Association which can provide you with a virtual assistant. Through this aide, teachers can become productive in class when errands like editing PowerPoint slides and keying in details into the grading software are taken care of by a Virtual Assistant.

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- **Todoist's**: Todoist's tagline is "Free up your mental space," and it does just that. It lets you take control of your day with a customizable to-do list and tasks. It helps you manage your jobs by allowing you to **color-code tasks** according to priority

- **Wunderlist**: Wunderlist is an easy-to-use and free task manager that can help teachers and other professionals make and organize lists. You can use it on your computer, tablet, and phone at no cost, a great feature for teachers. Wunderlist is very easy to use and you can take your lists with you everywhere you go so that you are working smarter, not harder.

- **Edmodo**: Edmodo provides the learning community with unlimited resources. It is an assessment tool that helps educators build interactive classrooms with polls, quizzes, notes, questions, and assignments.

- **Groupware/Online collaboration tools** is a class of computer programs that enables individuals to collaborate on projects with a common goal from geographically dispersed locations through shared Internet interfaces as a means to communicate within the group. Eg. • Google Workspace, Microsoft 365, Airtable, Asana, clickup,Microsoft teams, etc

Health issues related to use of ICT Tools

Generally a hazard is a situation in the workplace that has the potential to harm the health and safety of people or to damage plant and equipment. Common hazards encountered by computer technicians and users includes; Physical Hazards, Chemical Hazards and biological hazards

Health Hazards/Ergonomic Hazards

An ergonomic hazard is a physical factor within the environment that harms the musculoskeletal system (Especially in the workplace). With increasing number of computer users, the number health issues are also growing at a rapid pace in the work place. Some of the computer related physical injuries includes:

1) Musculoskeletal problems/Back problems

Many computer users suffer serious back problems. This is probably due to a poor posture or an awkward position while sitting at a computer.

Practical tips

- A fully adjustable chair should avoid poor posture.
- Footrests can reduce these problems.
- Screens should tilt and turn to a position that avoids awkward movements.

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2) Repetitive strain injury:

Repetitive Strain Injury (RSI) is damage to the fingers, wrists and other parts of the body due to repeated movements over a long period of time.

Practical Tips:

- Typing should be gentle. Do not fix your wrists in a certain position while typing.
- Have a five-minute break from typing every hour.
- Relax your arms or stretch when you are not typing or using the mouse.

3) Eye Strain

Eyes can become strained after staring at a computer screen for a long time, particularly if working in bad light, in glare or with a flickering screen.

Practical tips

- Adjust the contrast and brightness of your computer screen such that your eyes are not strained.
- Maintain a proper distance from the screen.
- Use screens that do not flicker.
- Take regular breaks - do not work for more than one hour without a break.

4) Headache

Because of increased muscle tension or pain in the neck at the base of the skull, headache is common problem with computer use. Many a times, prolonged use can affect eye power which needs vision correction. This can also result in headache.

Practical tips

- Get your eye power tested if headache is a common problem with computer use.
- Take breaks in between and circle your neck a bit to release the strain.

5) Obesity

Studies have shown that prolonged use of computers, especially in children, is the major contributing factor of sedentary lifestyle and childhood obesity.

Practical tips:

- Set limits for your children if they insist to playing computer games non-stop.
- Encourage your children to play outdoor games or to take up a hobby. Involve them in extra-curricular activities
- Adults who work for 7-8 hours should avoid spending time on computer after reaching home. Your body and mind both need to relax.

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6) Stress disorders

Prolonged computer use along with other factors like poor health, work pressure and job environment can make you susceptible to stress. Moreover, the longer you uphold the stress, the more susceptible you become to other health issues mentioned above. It can also lead to loss of concentration, dizziness and weariness.

Practical tips:

- Be proactive and take necessary measures to beat stress before it affects your health.

Chemical hazards

Exist in the form of gases, dust, corrosives, vapor and liquids. These chemicals can become dangerous to persons, when inhaled or absorbed through the skin. -Can cause breathing problems, skin irritation or burns.

- When you open a computer you will find circuit boards, and circuit boards often contain poisonous metals that are used in the manufacturing process, including mercury and lead. Both of these can have profound health effects in humans. Mercury poisoning causes damage to the central nervous system, liver, and other organs and causes impairment of the senses (vision, speech, and hearing). Lead toxicity can cause anemia, irreversible neurological damage, cardiovascular effects, gastrointestinal symptoms, and renal disease.
- Computer screens (CRT) also contain phosphors and barium compounds, which are toxic, and the glass may be leaded. Liquid crystal display screens may also contain lead, as well as copper at levels in excess of regulatory limits. The backlight may also contain mercury.
- Some UPS devices and some older computers use lead acid batteries (like car batteries). These contain sulfuric acid that's very corrosive and can pose a big danger if they leak.
- Laptop Batteries also contain heavy metals, including lead, mercury, nickel, and cadmium which are also considered to be hazardous.
- Ozone irritation: Health experts have suggested that ozone emitted from laser printers can lead to breathing problems.

Controlling chemical hazards in the workplace

- Reduce or eliminate the use of hazardous chemicals whenever possible.
- Maintain adequate ventilation systems to reduce concentrations of airborne chemicals.
- Practicing good personal hygiene (e.g. washing hands) and maintaining regular workplace cleaning routines can reduce the amount of a chemical substance that is absorbed by a worker's body.
- Introduce administrative controls to minimize exposure to chemicals (e.g. rotate workers through different jobs or locations, perform maintenance work in off-hours so that accidental release of toxic substances will affect fewer workers).

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- Use personal protective equipment and devices.
- Maintain equipment in good order to prevent leaks and breakdowns that may release toxic substances.

Physical Hazards may include

High temperatures: E.g. Heat arising from computer components such as the processor and the power supply unit can cause fire in to the office if not taken care of. **Noise:** In most cases, noise in the office environment (e.g. office equipment, ringing phones, air conditioning/fans, printers) is unlikely to reach levels that are hazardous to hearing. The **postures** and practices a person adopts throughout the day can have a significant impact on their risk of developing computer-related health issues. The **health** hazards as discussed above are also be considered as physical hazards arising from the use of computers.

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DATA SECURITY AND CONTROL

Data & Information must be protected against unauthorized access, disclosure, modification or damage. ***Data security*** is the protection of data & information from accidental or intentional disclosure to unauthorized persons.

SECURITY THREATS TO DATA & INFORMATION

1). COMPUTER VIRUSES

A computer virus is a destructive program that attaches itself to other files when the files are opened for use, and installs itself on the computer, without the knowledge of the user.

A computer virus is a program designed specifically to damage other programs or interfere with the proper functioning of the computer system.

Types of computer viruses.

1. *Boot sector viruses* – they destroy the booting information on storage devices.

2. *File viruses* – they attach themselves to files either erasing or modifying them.

3. *Hoax viruses* – they come as e-mails with an attractive subject & activate themselves when the e-mail is opened.

4. *Trojans* – they appear to perform necessary functions, but perform other undesirable activities in the background without the knowledge of the user.

5. *Worms* – viruses that stick in the computer memory.

6. *Backdoors* – may be a Trojan or Worm that allows hidden access to a computer system.

Types of destructions/damages caused by a virus attack

- Delete or modify data, information & files on storage devices (disks) or memory during normal program execution
- Systematically destroy all the data in the computer memory.
- Lock the keyboard.
- Uses up computer memory/space, hence slowing down its performance
- Can change keystroke values
- Cause boot failure.

Sources of viruses

a) ***Contact with contaminated systems:*** If a disk is used on a virus infected computer, it could become contaminated. If the same diskette is used on another computer, then the virus will spread.

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b) ***Use of pirated software:*** Pirated software may be contaminated by a virus code or it may have been amended to perform some destructive functions which may affect your computer.

c) ***Infected proprietary software:*** A virus could be introduced when the software is being developed in laboratories, and then copied onto diskettes containing the finished software product.

d) ***Fake games:*** Some virus programs behave like games software. Since many people like playing games on computers, the virus can spread very fast.

f) ***Updates of software distributed via networks:*** Viruses programs can be spread through software distributed via networks.

Symptoms of viruses in a computer system

The following symptoms indicate the presence of a virus in your computer:

- Boot failure.
- Files & programs disappearing mysteriously.
- Unfamiliar graphics or messages appearing on the screen,
- Slow booting.
- Gradual filing of the free space on the hard disk.
- Corruption of files and programs.
- Programs taking longer than usual to load.
- Unusual error messages occurring more frequently.

Control measures against viruses.

- Install up-to-date (or the latest) antivirus software on the computers.
- Restrict the movement of foreign storage media, e.g., flash disks in the computer room.
- Scan all storage media e.g. flash disks for viruses before using them.
- Avoid opening mail attachments before scanning them for viruses.
- Backup all software & data files at regular intervals.
- Avoid pirated software. If possible, use the software from the major software houses.

2). ***UNAUTHORIZED ACCESS***

Data & information is always under constant threat from people who may want to access it without permission. Such persons will usually have a bad intention, either to commit fraud, steal the information & destroy or corrupt the data.

Unauthorized access may take the following forms:

a). ***Eavesdropping:*** This is tapping into communication channels to get information

b). ***Surveillance (monitoring):*** This is where a person may monitor all computer activities done by another person or people.

c). ***Industrial espionage:***

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Industrial espionage involves spying on a competitor so as to get or steal information that can be used to finish the competitor or for commercial gain.

d). an employee who is not supposed to see some sensitive data gets it, either by mistake or design.

e). Forced entry into the computer room through weak access points.

Control measures against unauthorized access.

i). Enforce data & information access control policies on all employees to control access to data.
ii). Keep the computer room closed when nobody is using it.
iii). Reinforce weak access points, e.g., doors & windows with metallic grills & burglar alarms.
iv). Use file passwords to prevent any person from getting access to the electronic files.
v). Enforce network security measures, e.g., use of firewalls.
vi). Encrypt the data & information during transmission.
vii). Perform frequent Audit trails to identify threats to data & information.

3). *COMPUTER ERRORS & ACCIDENTAL ACCESS*
Errors and accidental access to data & information may be as a result of:

- Mistakes made by people, e.g., one may print sensitive reports & unsuspectingly give them to unauthorized persons.

- People experimenting with features they are not familiar with. E.g., a person may innocently download a file without knowing that it is self-installing or it may be dangerous to the system.

Control measures against computer errors & accidents.

i). Restrict file access to the end-users and technical staff in the organization, i.e., deny access of certain files & computers to certain groups of end-users.

ii). Set up a comprehensive error-recovery strategy in the organization.

4). *THEFT*
The threat of theft of data & information, hardware & software is real. Some information is so valuable such that business competitors or some governments can decide to pay somebody a fortune so as to steal the information for them to use.

Control measures against theft of information, hardware, & software.

i). Create backups & store them in locations away from the main computing center.
ii). Reinforce weak access points, e.g., the windows, doors, & roofing with metallic grills and strong padlocks.
iv). Employ guards to keep watch over data & information centres and backups.

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COMPUTER CRIMES

A **computer crime** is a deliberate theft or criminal destruction of computerized data. It is the use of computer hardware, software, or data for illegal activities, e.g., stealing, forgery, defrauding, etc.

Types of computer crimes

The following are the major types of computer crimes:

1. Trespass.
2. Hacking.
3. Tapping.
4. Cracking.
5. Piracy.
6. Fraud (Theft of money)
7. Sabotage.
8. Alteration of data.
9. Theft of computer time / Theft of service.
10. Theft of data, information or programs.
11. Damage of software.

Trespass.

- **Trespass** refers to the illegal physical entry to restricted places where computer hardware, software & backed up data is kept.

- It can also refer to the act of accessing information illegally on a local or remote computer over a network.

Hacking

Hacking is an attempt to invade the privacy of a system, either by tapping messages being transmitted along a public telephone line, or through breaking security codes & passwords to gain unauthorized entry to the system data and information files in a computer.

Reasons for hacking

- To copy or corrupt the information.
- As a hobby to test their expertise.
- Some do it for computer & software producing companies that want to secure their systems by reducing weaknesses discovered after professional hacking.

Hacking is done by skilled programmers referred to as **Hackers**. **Hacker** is a person who gains unauthorized access to a computer network for profit, criminal mischief, or personal gain.

Tapping

Tapping involves listening to a transmission line to gain a copy of the message being transmitted.

Cracking

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Cracking is the use of guesswork by a person trying to look for a weakness in the security codes of a software in order to get access to data & information.

Piracy

Piracy means making illegal copies of copyrighted software, data, or information either for personal use or for re-sale.

Ways of reducing piracy:

i) Enact & enforce copyright laws that protect the owners of data & information against piracy.
ii) Make software cheap enough to increase affordability.
iii) Use licenses and certificates of authenticity to identify originals.
iv) Set installation passwords that prevent illegal installation of software.

Fraud

Fraud is the use of computers to conceal information or cheat other people with the intention of gaining money or information. Fraud may take the following forms:

a). *Input manipulation:* Data input clerks can manipulate input transactions, e.g., they can create dummy (ghost) employees on the Salary file or a ghost supplier on the Purchases file.

b). *Production & use of fake documents:* E.g., a person created an intelligent program in the Tax department that could credit his account with cents from all the tax payers. He ended up becoming very rich before he was discovered.
Fraudsters can either be employees in the company or outsiders who are smart enough to defraud unsuspecting people.

Reasons that may lead to computer fraud

- For economic gain (i.e., to gain money or information).
- To gain respect (self-worth)

Security measures to prevent fraud:

i) Careful recruitment of staff.
ii) Set up a clear & firm management policy on crimes & frauds.
iii) Restrict access to computer room or terminal.
iv) Use transaction & fill logs to monitor access to sensitive areas of the system.
v) Monitor & investigate error logs and reports on regular basis.
vi) Carry out risk analysis to examine the exposure of the organization to possible fraud.

Sabotage

Sabotage is the illegal or malicious destruction of the system, data or information by employees or other people with grudges with the aim of crippling service delivery or causing great loss to an

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organization. Sabotage is usually carried out by discontented employees or those sent by competitors to cause harm to the organization.

The following are some acts of saboteurs which can result in great damage to the computer centres:
- Using Magnets to mix up (mess up) codes on tapes.
- Planting of bombs.
- Cutting of communication lines.

Alteration

Alteration is the illegal changing of stored data & information without permission with the aim of gaining or misinforming the authorized users.
Alteration is usually done by those people who wish to hide the truth. It makes the data irrelevant and unreliable.

Alteration may take place through the following ways:

a). *Program alteration:*
This is done by people with excellent programming skills. They do this out of malice or they may liaise with others for selfish gains.

b). *Alteration of data in a database:*
This is normally done by authorized database users, e.g., one can adjust prices on Invoices, increase prices on selling products, etc, and then pocket the surplus amounts.

Security measures to prevent alteration:
i) Do not give data editing capabilities to anybody without vetting.
ii) The person altering the data may be forced to sign in order for the system to accept altering the information.

Theft of computer time.

Employees may use the computers of an organization to do their own work, e.g., they may produce publications for selling using the computers of the company.

Theft of data (i.e., commercial espionage).

Employees steal sensitive information or copy packages and sell them to outsiders or competitors for profit. This may lead to a leakage of important information, e.g., information on marketing strategies used by the organization, research information, or medical reports.

DETECTION & PROTECTION AGAINST COMPUTER CRIMES

The following measures can be taken to detect & prevent computer crimes, and also seal security loopholes.

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Audit trails

This is a careful study of an information system by experts in order to establish (or, find out) all the weaknesses in the system that could lead to security threats or act as weak access points for criminals.

Data encryption

Data being transmitted over a network faces the dangers of being tapped, listened to, or copied to unauthorized destinations. To protect such data, it is mixed up into a form that only the sender & the receiver can be able to understand by reconstructing the original message from the mix. This is called **Data encryption**.

Log files

These are special system files that keep a record (log) of events on the use of the computers and resources of the information system.
Each user is usually assigned a *username* & *password* or account. The information system administrator can therefore easily track who accessed the system, when and what they did on the system.

Firewalls

A **Firewall** is a device or software system that filters the data & information exchanged between different networks by monitoring & controlling access to or from protected networks.

Environmental threats to computers & Information systems.

1). Fire.

Fire destroys data, information, software & hardware.
Security measures against fire:
- Use fire-proof cabinets & lockable metal boxes for floppy disks.
- Use of backups.
- Install firefighting equipment, e.g., fire extinguishers.
- Have some detectors.
- Training of fire-fighting officers.
- Observe safety procedures, e.g., avoid smoking in the computer rooms.
- Have well placed exit signs.
- Contingency plans.

2). Water, floods & moisture.

This causes rusting of the metallic components of the computer.
Security measures against water, floods & moisture:
- Set up computer rooms on higher grounds to avoid floods & humidity.
- Avoid installing computer components in the basement.
- There should be adequate drainage system.
- Use water-proof ceilings & floors.

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3). Lightening, electricity & electrical storms.
This causes power failure that can cause damage to data, which has not been transferred to permanent storage devices.

Security measures:
- Install facilities to control power fluctuations, e.g., use of Uninterrupted power source (UPS)
- Use power stabilizers.
- Have standby power generators/sources.
- Have lightening arresters in the building.

4). Excessive Heat or Temperature.
Excessive heat or temperature from the computer itself or from the surrounding environment can destroy computer storage media or devices.

Security measures:
- There should be efficient ventilation system.
- Use a cooling system in the computer rooms, e.g., cooling fans & air conditioners.

5). Computer virus attack.
A **virus** is a rogue software program that spreads rampantly through computer systems, destroying data or causing the system to break down.

Security measures against computer virus:
- Make backup copies of software, and store the copies off-site.
- Restrict access to programs & data on a 'need-to-use' basis.
- Check all programs regularly for change of size, as this could be a sign of virus infiltration.
- Be careful with 'Shareware' and 'Freeware' programs, as they are the major entry points for viruses.
- Make sure all purchased software is in its original sealed-disk containers.

6). Smoke and Dust.
Dust and Smoke particles settle on storage devices and may scratch them during Read/write operation.
Security measures:
- Have dust mats or carpets to prevent entry of dust.
- Fit the computer room with special Curtains to reduce entry of dust particles.
- Cover the devices with Dust covers when cleaning the room.
- Remove shoes before entering the room to prevent dust.

7). Terrorist attack.
This includes activities such as:

Political terrorists
Criminal type of activities

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Individuals with grudges, or
People intending to cause general destruction.

Security measures:
- Hiring of security guards to control physical access to the building housing the computer room.
- Activities that can cause terrorism should be avoided, e.g., exploitation of workers.
- Have double door & monitoring devices.
- Use of policies.
- System auditing / use of log files.
- Use of passwords.
- Punitive measures.
- Encryption of data.
- Use of firewalls.
- Consult & co-operate with the Police and Fire authorities on potential risks.

8). People.
People threats include:
- Carelessness.
- Clumsiness.
- Accidental deletion of data, information or programs.
- Vandalism, i.e., theft or destruction of data, information or programs & hardware.
- Piracy of copyrighted data & software.

Security measures against Carelessness & Clumsiness:
- Better selection of personnel.
- Have a good office layout.
- Improve employee training and education.
- Limit access to data and computers.
- Regular backups.
- Use of Undelete & Unformat utilities.

Security measures against Vandalism:
- Should have a sensitive attitude to office behaviour.
- Tighten security measures, e.g., install alarm systems, burglar-proof doors/windows, & roofs).
- Limit access to sensitive company information.
- Use Keyboard lock on terminals used by authorised users.
- Use of disk locks.
- Punitive measures.

Assignment 1: Discuss ethical and legislation concerns related to ICT in learning

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